

4.14

$$E(X) = 1000 = \frac{1}{\lambda} \rightarrow \lambda = 0.001$$

$$X \sim \text{Exp}(0.001)$$

a

$$\begin{aligned} P(X > 2000) &= \int_{2000}^{\infty} 0.001e^{-0.001x} dx \\ &= e^{-0.001 \cdot 2000} - 0 = e^{-2} \end{aligned}$$

b

$$\begin{aligned} P(X > 2000 | X > 500) &= \frac{P(X > 2000 \cap X > 500)}{P(X > 500)} = \frac{P(X > 2000)}{P(X > 500)} \\ P(X > 500) &= \int_{500}^{\infty} 0.001e^{-0.001x} dx \\ &= e^{-0.001 \cdot 500} - 0 = e^{-1/2} \\ P(X > 2000 | X > 500) &= \frac{e^{-2}}{e^{-1/2}} = e^{-3/2} \end{aligned}$$

4.49

Let X denote how long a unit lasts (in years). We know that $X \sim \text{Exp}(\frac{1}{10})$.

Our goal is to find (C, r) s.t. $0 = (C + 200) \times P(X > r) + (C - 600) \times P(X < r)$

$$\rightarrow (C + 200) \int_r^{\infty} \frac{1}{10} e^{-\frac{x}{10}} dx + (C - 600) \int_0^r \frac{1}{10} e^{-\frac{x}{10}} dx$$

$$\rightarrow (C + 200)(e^{-\frac{r}{10}}) + (C - 600)(1 - e^{-\frac{r}{10}})$$

$$\rightarrow C - 600 + 800e^{-\frac{r}{10}} = 0$$

$$C \text{ as a function of } r: C = 600 - 800e^{-\frac{r}{10}}$$

Thus, pairs $(C = 600 - 800e^{-\frac{r}{10}}, r = r)$ result in a profit of 0.

For $r = 2$, to result in 0 profit then $C = -54.98$ (I give the warranty holder ~\$50). Since I am expecting a profit of 200 dollars from people who do not buy the warranty I would price the warranty at a similar profit threshold and say the price for the full covered warranty is $C = 150$.

5.4

a

$$M_X(t) = e^{6t^2} \text{ when } |t| < 2$$

$X \sim N(0, 12)$ and bound by $|t| < 2$

b

$$M_Y(t) = \frac{2}{2-t} \text{ for } t < 0.5$$

$Y \sim \text{Exp}(2)$ and $t < 0.5$

c

$$M_Z(t) = \infty \text{ for } t \geq 5$$

Z could follow an exponential distribution with $\lambda \leq 5$

Thus we can determine Z as an exponential distribution but we cannot determine λ

d

$$M_W(2) = 2$$

$W \sim \text{Exp}(4)$ because for exponential distribution with $\lambda = 4$, $M(2) = \frac{4}{4-2} = 2$

Alternatively W could follow $P(W = 0) = 1/2$ and $P(W = \frac{\ln 3}{2}) = 1/2$

$$\text{Thus } M_W(2) = \sum_i e^{2i} P(W = i) \rightarrow 1/2 e^{2 \cdot 0} + 1/2 e^{2 \cdot \frac{\ln 3}{2}} = 1/2 + 3/2 = 2$$

Therefore not sufficient information to determine the distribution of W .

5.8

$$f_x = \frac{1}{b-a} \text{ if } x \in [a, b] \text{ and } 0 \text{ else}$$

$$f_x = \frac{1}{3} \text{ if } x \in [-1, 2] \text{ and } 0 \text{ else}$$

$$f_y(y) = \frac{f_x(g^{-1}(y))}{g'(g^{-1}(y))}$$

$$g(x) = x^2$$

$$g^{-1}(y) = \sqrt{y}$$

$$g'(x) = 2x$$

$$f_y(y) = \frac{1/3}{2\sqrt{y}} = \frac{1}{6\sqrt{y}}$$

Now the bounds of Y are not simply $-1^2, 2^2$ because this results in $[1, 4]$ which does not include values of $x \in (-1, 1)$. We can think about it as $x \in [-1, 1]$ maps to values of $y \in [0, 1]$, meaning this section is mapped twice and thus follows $\frac{1}{3\sqrt{y}}$

$$f_y(y) = \begin{cases} \frac{1}{3\sqrt{y}} & \text{if } x \in [0, 1) \\ \frac{1}{6\sqrt{y}} & \text{if } x \in [1, 4] \\ 0 & \text{else} \end{cases}$$

5.11

MORE CHECK CANVAS FOR UPDATES